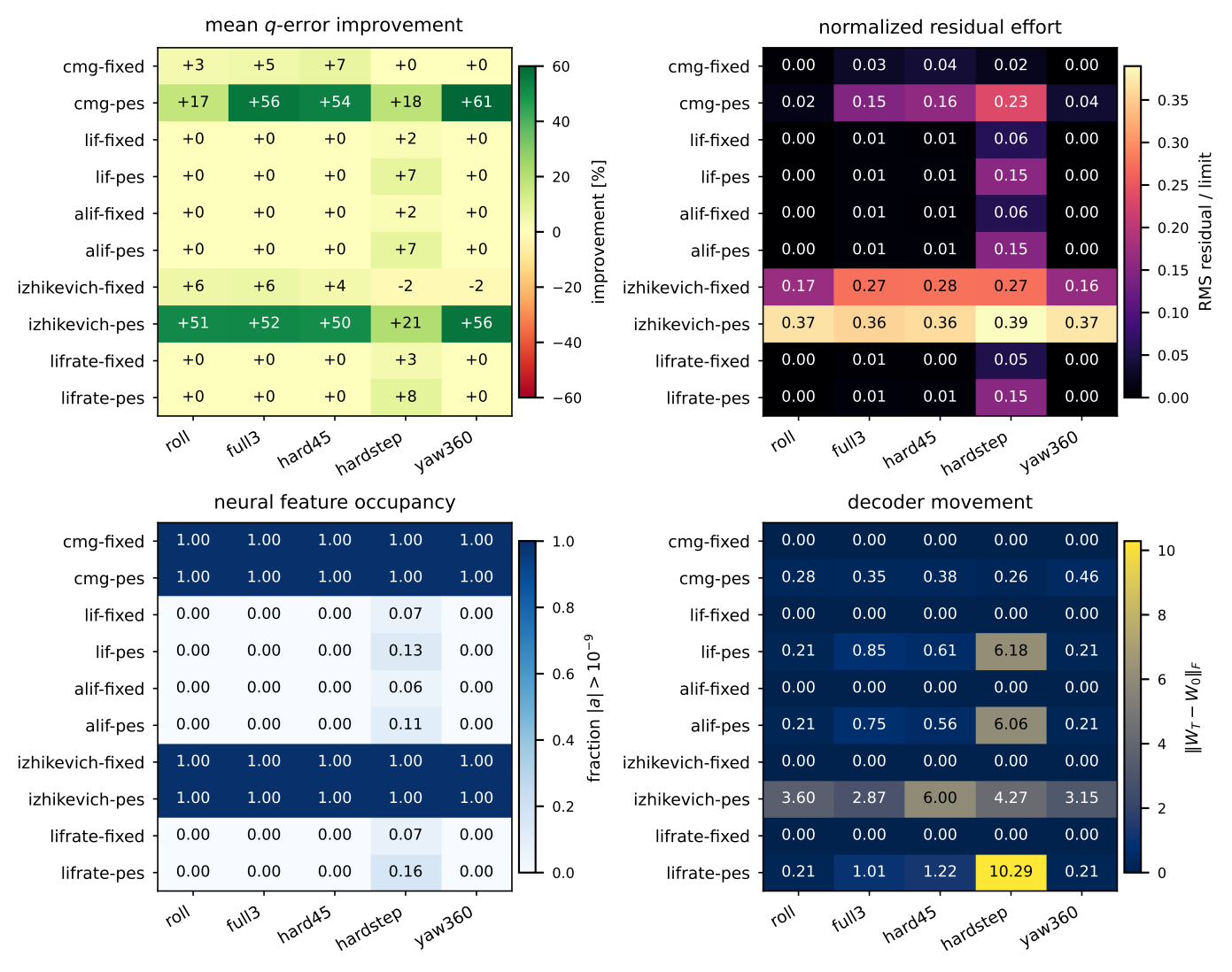
mean q-error improvement
cmg-fixed +3 +5 +7 +0 +0
cmg-pes +17 +56 +54 +18 +61
lif-fixed +0 +0 +0 +2 +0
lif-pes +0 +0 +0 +7 +0
alif-fixed +0 +0 +0 +2 +0
alif-pes +0 +0 +0 +7 +0
izhikevich-fixed +6 +6 +4 -2 -2
izhikevich-pes +51 +52 +50 +21 +56
lifrate-fixed +0 +0 +0 +3 +0
lifrate-pes +0 +0 +0 +8 +0
roll full3 hard45 hardstep yaw360
60 40 20 0 −20 −40 −60
improvement [%]
normalized residual effort
cmg-fixed 0.00 0.03 0.04 0.02 0.00
cmg-pes 0.02 0.15 0.16 0.23 0.04
lif-fixed 0.00 0.01 0.01 0.06 0.00
lif-pes 0.00 0.01 0.01 0.15 0.00
alif-fixed 0.00 0.01 0.01 0.06 0.00
alif-pes 0.00 0.01 0.01 0.15 0.00
izhikevich-fixed 0.17 0.27 0.28 0.27 0.16
izhikevich-pes 0.37 0.36 0.36 0.39 0.37
lifrate-fixed 0.00 0.01 0.00 0.05 0.00
lifrate-pes 0.00 0.01 0.01 0.15 0.00
roll full3 hard45 hardstep yaw360
0.35 0.30 0.25 0.20 0.15 0.10 0.05 0.00
RMS residual / limit
neural feature occupancy
cmg-fixed 1.00 1.00 1.00 1.00 1.00
cmg-pes 1.00 1.00 1.00 1.00 1.00
lif-fixed 0.00 0.00 0.00 0.07 0.00
lif-pes 0.00 0.00 0.00 0.13 0.00
alif-fixed 0.00 0.00 0.00 0.06 0.00
alif-pes 0.00 0.00 0.00 0.11 0.00
izhikevich-fixed 1.00 1.00 1.00 1.00 1.00
izhikevich-pes 1.00 1.00 1.00 1.00 1.00
lifrate-fixed 0.00 0.00 0.00 0.07 0.00
lifrate-pes 0.00 0.00 0.00 0.16 0.00
roll full3 hard45 hardstep yaw360
1.0 0.8 0.6 0.4 0.2 0.0
fraction $|a| > 10^{-9}$
decoder movement
cmg-fixed 0.00 0.00 0.00 0.00 0.00
cmg-pes 0.28 0.35 0.38 0.26 0.46
lif-fixed 0.00 0.00 0.00 0.00 0.00
lif-pes 0.21 0.85 0.61 6.18 0.21
alif-fixed 0.00 0.00 0.00 0.00 0.00
alif-pes 0.21 0.75 0.56 6.06 0.21
izhikevich-fixed 0.00 0.00 0.00 0.00 0.00
izhikevich-pes 3.60 2.87 6.00 4.27 3.15
lifrate-fixed 0.00 0.00 0.00 0.00 0.00
lifrate-pes 0.21 1.01 1.22 10.29 0.21
roll full3 hard45 hardstep yaw360
10 8 6 4 2 0
$\|W_T - W_0\|_F$